Middle East Technical University
Department of Electrical and Electronics Engineering

EE201 – Circuit Theory I

Lecture 9: Trees, Cutsets and Fundamental Loops

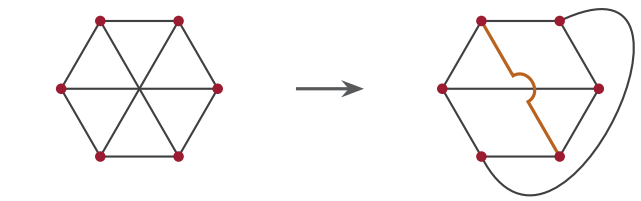
M. Mert Ankaralı

Colour code used in these notes and in the slides

current i | voltage v, cutsets, loops | tree branches | nodes ①

1. Beyond planar graphs

Meshes need a drawing with no crossings. A nonplanar graph has no such drawing, so it has no meshes and no mesh matrix. In the hexagon below, the three diagonals cross at the centre. Moving one diagonal outside removes one crossing, but a jumper wire (a hop) is still needed.



crossings at the centre jumper wire: still one hop

How do we then find $n-1$ independent KCL and $b-n+1$ independent KVL equations? This lecture gives a method that needs no planar drawing and works for **any connected graph**.

2. Trees

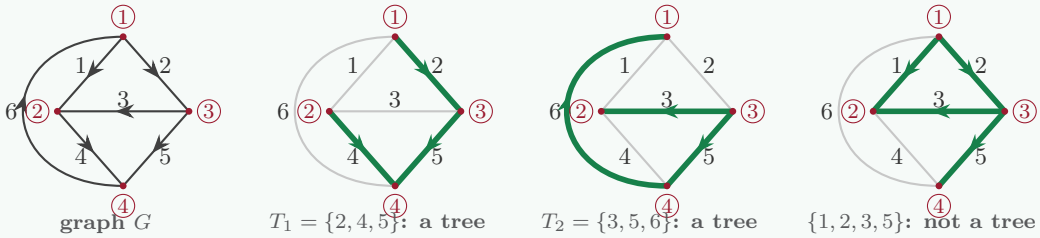
Tree

Let G be a connected graph. A subgraph T of G is a **tree** if

1. T contains all the nodes of G,
2. T is connected, and
3. T contains no loops.

Example 9.1 — trees of the bridge graph

Which of the green subgraphs are trees?



T_1 and T_2 reach all four nodes, are connected and have no loop. The set $\{1,2,3,5\}$ contains the loop of branches 1, 2, 3, so it is not a tree. A graph has many trees.

A tree has $n-1$ branches

Build the tree one branch at a time, starting from a single node. Each new branch must reach a new node: a branch between two nodes already reached would close a loop. All n nodes must be reached, so the tree has exactly $n-1$ branches. The other $b-n+1$ branches are left out.

3. Cutsets

Cutset

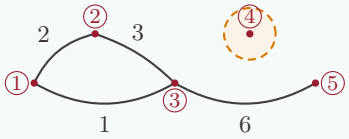
Let G be a connected graph. A set C of branches is a **cutset** if

1. removing all the branches of C leaves the graph unconnected, and
2. removing all the branches of C except any one of them leaves it connected.

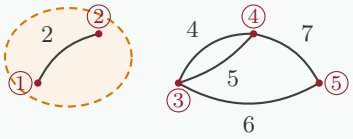
The second condition makes the cutset minimal: every branch in it is needed. Minimal does not mean fewest branches; cutsets of the same graph can have different sizes.

Example 9.2 — cutsets of a figure-eight graph

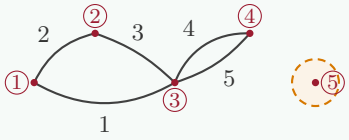
In each panel the removed branches are deleted; look at what is left.



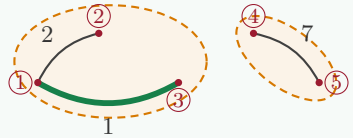
$C_1 = \{4, 5, 7\}$: node ④ cut off



$C_2 = \{1, 3\}$: nodes ①, ② cut off



$C_3 = \{6, 7\}$: node ⑤ cut off



$\{1, 3, 4, 5, 6\}$ with 1 put back: still two pieces

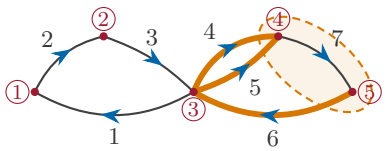
For C_1: removing branches 4, 5, 7 cuts off node ④, so condition 1 holds. Putting back any one of them reaches node ④ again, so condition 2 holds. C_2 and C_3 are checked the same way. With all branches back, the dashed closed surface around the cut-off piece crosses exactly the branches of the cutset.

The set $\{1, 3, 4, 5, 6\}$ is *not* a cutset. Removing it leaves three pieces. With branch 1 put back, the graph is still unconnected, so condition 2 fails. Branches 1 and 3 are not needed: $\{4, 5, 6\}$ alone is a cutset.

The set $\{4, 6\}$ is not a cutset either, but for the opposite reason. Removing it leaves the graph connected, since branch 5 still joins nodes ③ and ④, so condition 1 fails. Branches 4 and 5 are parallel: both join ③ and ④. A closed surface that crosses one of them crosses the other too. So parallel branches are in a cutset together or not at all.

KCL on a cutset. A cutset splits the graph into two parts. A closed surface around one part crosses exactly the branches of the cutset. So by KCL on that closed surface, the currents of the cutset branches add up to zero (each with + if it leaves the surface, − if it enters).

Take the branch currents below, in the arrow directions. They satisfy KCL at every node. Consider the cutset $\{4, 5, 6\}$, with the surface around nodes ④ and ⑤.



branch	1	2	3	4	5	6	7
current (A)	2	2	2	1	2	3	3

Current i_6 leaves the surface; i_4 and i_5 enter it:

$$i_6 - i_4 - i_5 = 3 - 1 - 2 = 0.$$

This is the sum of KCL at nodes ④ and ⑤: $(i_7 - i_4 - i_5) + (i_6 - i_7) = 0$. The current i_7 is inside the surface and cancels.

4. Fundamental cutsets

Fix a tree T of a connected graph G. The branches of G that are not in T form the **co-tree**. Now pick one tree branch β and remove it. The tree falls into two parts. The branches of G that join these two parts form a cutset.

Fundamental cutset

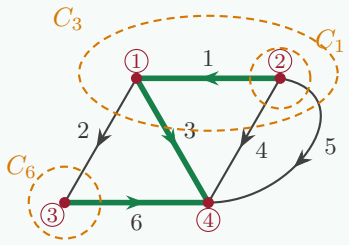
The **fundamental cutset** of a tree branch β is

- β plus the co-tree branches that join the two parts of the tree;
- it contains only one tree branch, β;
- its direction is that of β.

It has only one tree branch: every other tree branch lies inside one of the two parts, so it does not join them. For the direction, draw the closed surface around one part. If β leaves the surface, the cutset points out of the surface; if β enters, it points in. In KCL a branch counts $+$ if it crosses the surface the same way as β, and $-$ otherwise.

Example 9.3 — fundamental cutsets and KCL

The graph G has $n = 4$ nodes and $b = 6$ branches. Take the tree $\{1, 3, 6\}$ (green, $n - 1 = 3$ branches). The co-tree is $\{2, 4, 5\}$ ($b - n + 1 = 3$ branches).



tree br.	cutset	KCL
1	$\{1, 4, 5\}$	$i_1 + i_4 + i_5 = 0$
3	$\{3, 2, 4, 5\}$	$i_3 + i_2 + i_4 + i_5 = 0$
6	$\{6, 2\}$	$i_6 - i_2 = 0$

Each surface encloses one of the two parts left after removing a tree branch. For C_3: removing branch 3 leaves the parts $\{①, ②\}$ and $\{③, ④\}$. Branches 3, 2, 4, 5 join them, and all four go from the upper part to the lower part, the same way as branch 3: every term has $+$.

KCL on the fundamental cutsets

A tree gives $n - 1$ fundamental cutsets, so $n - 1$ KCL equations. They are independent: each equation contains its own tree branch current, which appears in no other equation.

Remark: co-tree currents are enough. Suppose only the co-tree currents are known. Each fundamental cutset has one tree branch, so its KCL can be solved for that tree branch current. For G:

$$i_1 = -i_4 - i_5, \quad i_3 = -i_2 - i_4 - i_5, \quad i_6 = i_2.$$

For example, $i_2 = 1$, $i_4 = 1$, $i_5 = 2$ A give $i_1 = -3$, $i_3 = -4$, $i_6 = 1$ A. So the $b - n + 1$ co-tree currents fix every branch current, like mesh currents.

5. Fundamental loops

Fix a tree T of a connected graph G and pick one co-tree branch. Adding it to the tree closes exactly one loop: the co-tree branch plus the tree path between its two ends.

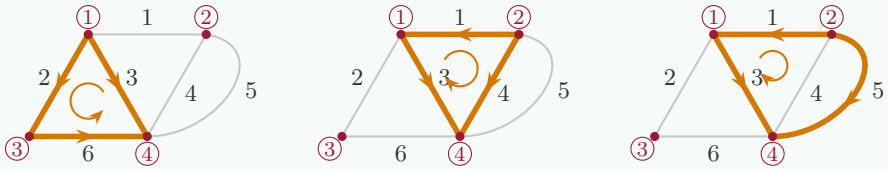
Fundamental loop

The **fundamental loop** of a co-tree branch is

- that co-tree branch plus the tree path between its ends;
- it contains only one co-tree branch;
- its direction is that of the co-tree branch.

Example 9.4 — fundamental loops and KVL

Same graph and tree as in Example 9.3. Each co-tree branch closes one loop. Walk it in the direction of its co-tree branch: a branch voltage counts + if its arrow points along the walk, − if against.



loop of 2: $\{2,6,3\}$ $v_2 + v_6 - v_3 = 0$

loop of 4: $\{4,3,1\}$ $v_4 - v_3 - v_1 = 0$

loop of 5: $\{5,3,1\}$ $v_5 - v_3 - v_1 = 0$

KVL on the fundamental loops

A tree gives $b-n+1$ fundamental loops, so $b-n+1$ KVL equations. They are independent: each equation contains its own co-tree branch voltage, which appears in no other equation.

Remark: tree voltages are enough. Suppose only the tree voltages are known. Each fundamental loop has one co-tree branch, so its KVL can be solved for that co-tree branch voltage. For G:

$$v_2 = v_3 - v_6, \quad v_4 = v_1 + v_3, \quad v_5 = v_1 + v_3.$$

For example, $v_1 = 2$, $v_3 = 3$, $v_6 = 1$ V give $v_2 = 2$, $v_4 = v_5 = 5$ V. So the $n-1$ tree voltages fix every branch voltage, like node voltages.

6. Matrix form

The fundamental cutset equations can be collected in a matrix, like the incidence matrix. The **fundamental cutset matrix** Q has one row per tree branch and one column per branch. An entry is $+1$ or -1 if the branch is in the cutset, in the same or the opposite direction, and 0 otherwise. The **fundamental loop matrix** B has one row per co-tree branch; an entry is $+1$ or -1 if the branch is in the loop, along or against its direction. For G, with tree $\{1,3,6\}$:

$$Q = \begin{array}{c} \\ 1 \\ 3 \\ 6 \end{array}\!\!\begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{array} \qquad B = \begin{array}{c} \\ 2 \\ 4 \\ 5 \end{array}\!\!\begin{array}{c} \begin{array}{cccccc} 1 & 2 & 3 & 4 & 5 & 6 \end{array} \\ \begin{bmatrix} 0 & 1 & -1 & 0 & 0 & 1 \\ -1 & 0 & -1 & 1 & 0 & 0 \\ -1 & 0 & -1 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

KCL on the fundamental cutsets is $Q\mathbf{i} = \mathbf{0}$, and KVL on the fundamental loops is $B\mathbf{v} = \mathbf{0}$. The tree columns of Q and the co-tree columns of B (boxed) form identity matrices, so the rows are independent.

The transposes give the remarks above in one line each. With the co-tree currents $\mathbf{i}_C = [i_2\ i_4\ i_5]^T$ and the tree voltages $\mathbf{v}_T = [v_1\ v_3\ v_6]^T$,

$$\mathbf{i} = B^T\mathbf{i}_C, \qquad \mathbf{v} = Q^T\mathbf{v}_T.$$

The first satisfies KCL for any $\mathbf{i}_C$, the second satisfies KVL for any $\mathbf{v}_T$. Finally $QB^T = \mathbf{0}$: every row of Q is orthogonal to every row of B.

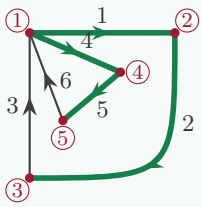
	incidence / mesh	fundamental cutset / loop
KCL	$A\mathbf{i} = \mathbf{0}$	$Q\mathbf{i} = \mathbf{0}$
KVL	$M\mathbf{v} = \mathbf{0}$	$B\mathbf{v} = \mathbf{0}$
$n-1$ voltages	$\mathbf{v} = A^T\mathbf{e}$ (node voltages)	$\mathbf{v} = Q^T\mathbf{v}_T$ (tree voltages)
$b-n+1$ currents	$\mathbf{i} = M^T\mathbf{i}_M$ (mesh currents)	$\mathbf{i} = B^T\mathbf{i}_C$ (co-tree currents)
orthogonality	$AM^T = \mathbf{0}$	$QB^T = \mathbf{0}$
works for	A: any; M: planar, nonseparable	any connected graph

7. Any connected graph

The tree is the key, and every connected graph has one. Its $n-1$ tree branches give the KCL equations, and its $b-n+1$ co-tree branches give the KVL equations. No drawing is needed. This works for nonplanar graphs, and it is also the fix for separable graphs met earlier.

Example 9.5 — the separable graph

Two loops joined only at node ①: $n = 5$, $b = 6$. Take the tree $\{1, 2, 4, 5\}$ (green, $n-1 = 4$). The co-tree is $\{3, 6\}$ ($b-n+1 = 2$).



tree br.	cutset	KCL
1	$\{1,3\}$	$i_1 - i_3 = 0$
2	$\{2,3\}$	$i_2 - i_3 = 0$
4	$\{4,6\}$	$i_4 - i_6 = 0$
5	$\{5,6\}$	$i_5 - i_6 = 0$

co-tree br.	KVL on its loop
3	$v_3 + v_1 + v_2 = 0$
6	$v_6 + v_4 + v_5 = 0$

The two fundamental loops are the outer loop and the inner triangle. The KCL equations give $i_1 = i_2 = i_3$ and $i_4 = i_5 = i_6$: one current for each loop. In total $4 + 2 = 6 = b$ independent equations. Node ① causes no trouble.

8. Looking ahead

A circuit has $2b$ branch unknowns. A tree reduces them: the $n-1$ **tree branch voltages** fix every branch voltage (like node voltages), and the $b-n+1$ **co-tree branch currents** fix every branch current (like mesh currents), for any connected graph.

Summary

- A **tree** of a connected graph contains all nodes, is connected and has no loop. It has $n-1$ branches; the other $b-n+1$ form the **co-tree**.
- A **cutset** is a minimal set of branches whose removal cuts the graph in two. KCL holds on every cutset: it is KCL on a closed surface.
- Each tree branch has a **fundamental cutset**: it and some co-tree branches. These give $n-1$ independent KCL equations. Knowing the co-tree currents gives all branch currents.
- Each co-tree branch has a **fundamental loop**: it and a tree path. These give $b-n+1$ independent KVL equations. Knowing the tree voltages gives all branch voltages.
- In matrix form: $Q\mathbf{i} = \mathbf{0}$, $B\mathbf{v} = \mathbf{0}$, $\mathbf{i} = B^T\mathbf{i}_C$, $\mathbf{v} = Q^T\mathbf{v}_T$, for any connected graph.

Reading

- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits* — trees, cutsets and fundamental loops.
- Lecture notes and videos of Prof. Emre Tuna, `users.metu.edu.tr/etuna/ee201` — notes pp. 17–18; Lecture 8 video: trees 16:10, cutsets 22:33, fundamental cutsets 30:35, fundamental loops 43:51.